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Subject : Fundamentals of Telecommunication Systems

Experiment Number : 02

Experiment Title : Overview and Demonstration of Mobile Network Components.

Date of Performance : 19/01/2026

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Experiment No. 02

Title: Overview and Demonstration of Mobile Network Components.

Aim:

To study and understand the architecture, major components, and basic working of a mobile communication network using free online demonstrations, animations, and educational videos.

Theory

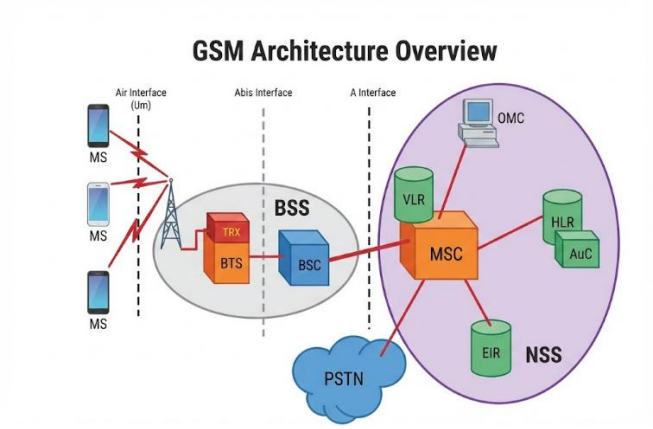
A mobile communication system enables wireless voice and data communication between users over a wide geographical area. Communication between the mobile handset and the network is established using radio waves, while the core network performs switching, routing, authentication, and mobility management.

A large service area is divided into small regions called cells, each served by a Base Transceiver Station (BTS). This cellular architecture allows frequency reuse, efficient spectrum utilization, and uninterrupted service to moving users.

A basic mobile network consists of three major subsystems:

- Mobile Station (MS) – user equipment
- Radio Access Network (RAN) – BTS and controllers
- Core Network (CN) – switching and subscriber databases

Block Diagram



Task 1: Identification and study of Mobile Network Components

Mobile Station (MS)

[Paste Screenshot / Draw Neat Sketch Here]

The Mobile Station is the user's mobile device that provides access to the cellular network. It converts voice into electrical and radio signals for wireless transmission. The MS communicates directly with the BTS using radio waves. It also performs encryption, decryption, and user authentication. Without the mobile station, a subscriber cannot initiate or receive calls.

Base Transceiver Station (BTS)

[Paste Screenshot / Draw Neat Sketch Here]

The BTS provides the radio interface between the mobile station and the network. It transmits and receives radio frequency signals within a defined cell area. The BTS handles modulation, demodulation, and signal amplification. It controls radio channels and signal strength. The BTS is essential for establishing wireless connectivity.

Base Station Controller (BSC)

[Paste Screenshot / Draw Neat Sketch Here]

The BSC controls and manages multiple BTS units in a given region. It allocates radio channels and manages handovers between cells. The BSC reduces the processing load on the core network. It ensures efficient utilization of radio resources. The BSC plays a key role in maintaining call quality.

Mobile Switching Center (MSC)

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The MSC is the central switching element of the mobile network. It manages call setup, routing, and termination. The MSC connects mobile calls to other mobile users and external networks like PSTN. It also handles mobility management and billing. The MSC acts as the heart of the core network.

Home Location Register (HLR)

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The HLR is a permanent database that stores subscriber information. It contains details such as subscriber identity, service profile, and current location. The HLR is used during authentication

and call routing. It ensures that only authorized users access the network. The HLR supports mobility across different regions.

Visitor Location Register (VLR)

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The VLR is a temporary database that stores information about subscribers currently visiting a service area. It works in coordination with the MSC and HLR. The VLR reduces signaling traffic to the HLR. It enables faster call setup for roaming users. The VLR improves network efficiency.

Task 2: Conceptual Working of a Mobile Call

Mobile Power ON and Network Registration

When the mobile phone is powered ON, it searches for nearby BTS signals. The network verifies subscriber identity using the HLR and VLR. After successful authentication, the mobile registers with the network. The user is now ready to make or receive calls.

Call Initiation by Mobile User

When the user dials a number, the mobile sends a call request to the BTS. The request is forwarded to the BSC and MSC. The MSC processes the dialed number. Call setup procedures are initiated.

Radio Link Establishment between MS and BTS

A dedicated radio channel is assigned between the mobile station and BTS. This channel carries voice and control signals. Stable communication is maintained throughout the call. Signal strength is continuously monitored.

Call Routing through BSC and MSC

The BSC forwards the call signaling to the MSC. The MSC determines the destination network. The call is routed to another mobile user or to PSTN/Internet. A complete communication path is established.

Handover between Cells (if the user moves)

When the user moves out of a cell, signal strength decreases. The BSC initiates a handover to a neighboring BTS. The call is transferred without interruption. This ensures continuous communication.

Call Termination

When the call ends, a disconnect signal is sent to the network. The MSC releases all allocated resources. The radio channel is freed. The call session is terminated completely.

Conclusion

This experiment demonstrates the working of a mobile communication system using cellular network architecture. The human voice is converted into electrical signals by the mobile device and transmitted as radio waves through base stations. These signals are switched, routed, and managed by the core network using subscriber databases. Finally, the received signals are reconstructed into sound at the destination, enabling reliable wireless communication.